and equally spaced **B1**

**2** **(a)** The rate of change of momentum of a body is directly proportional to the resultant force acting on the body and occurs in the direction of the resultant force. **B1**

**(b)** **(i)** $p = \Delta(mv)$

$= -65 \times 10^{-3} (5.2 + 3.7)$ **C1**

$= -0.58$ N s **A1**

**(ii)** $F = (0.58) / (7.5 \times 10^{-3}) = 77$ N **A1**

**(c)** **(i)** **1.** Force on the wall from the ball is equal to the force on ball from the wall **B1**

but in the opposite direction **B1**

(general statement of Newton's third law only score 1 mark)

**2.** momentum change of ball is equal **B1**

and opposite to momentum change of the wall

**OR** since there is no external resultant force acting on the system (ball and wall), the total momentum of the system is conserved

**OR** total change in momentum of ball and wall is zero

**(ii)** <u>kinetic</u> energy (of ball and wall) is reduced / not conserved, inelastic. **B1**

(allow relative speed of approach does not equal to relative speed of separation)